Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/05/2025
PI Name:	Walsh, Kenneth Ph.D.		
Project Title:	Space Radiation Exposure and Risk Mediated by Clonal Hematopoiesis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Charlottesville	**State:**	VA
Zip Code:	22903-3390	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	01/29/2021	**End Date:**	02/28/2027
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 02/28/2027 per the PI and NSSC documentation (Ed., 6/9/25).		
Key Personnel Changes/Previous PI:	1 Bachelor's candidate and 1 Bachelor's degree left the University/project. We have 1 Master's, 1 PhD, 1 postdoctoral fellow, and an MD/PhD student working in part on this project.		
COI Name (Institution):	Garrett-Bakelman, Francine M.D., Ph.D. (University of Virginia, Charlottesville) Hirschi, Karen Ph.D. (Yale University) Goukassian, David M.D., Ph.D. (ICAHN School of Medicine at Mount Sinai) Evans, Megan A (University of Virginia)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	During spaceflight, astronauts are exposed to many stresses that alter multiple physiological systems. The recent NASA Twins Study provided a highly detailed analysis of how prolonged, low orbit space travel may contribute to genotoxic stress, elevated DNA damage responses and genomic instability in leukocytes. The observed genomic instability during and after flight suggests that the ionizing radiation exposure caused DNA damage to hematopoietic stem cells that replenish blood cells throughout life. Thus, it is conceivable that these alterations will contribute to the development of hematologic malignancies and other chronic diseases through changes in immune cell function. Furthermore, these effects may be particularly magnified when traveling beyond Earth's geomagnetic field where there is increased exposure to high atomic number and high energy radiation. Recent epidemiological studies have documented the prevalence of somatic mutations within the cells of the hematopoietic system in healthy individuals. These acquired DNA mutations accumulate with age and, in some instances, can provide a competitive advantage to the mutant cell thus allowing for its clonal expansion. This phenomenon is known as clonal hematopoiesis of indeterminate potential (CHIP). While the mutational landscape of CHIP has only partially been deciphered, some of these clonal expansions can be attributed to somatic mutations in driver genes that are recurrently mutated in blood malignancies. These driver genes include epigenetic regulators (TET2, DNMT3A, ASXL1), spliceosome components (SF3B1, SRSF2), signaling proteins (JAK2), and DNA damage response molecules (TP53, PPM1D). Studies show that CHIP is associated with an increased risk of all-cause mortality. While there is a marked increase in the frequency of hematological cancer in individuals with CHIP, which is to be expected, the major cause of the increased mortality in these populations appears to be an increase in cardiovascular diseases including coronary heart disease, stroke, and early-onset myocardial infarction. Studies in the applicants' laboratories have provided evidence for a causal link between CHIP, derived from mutations in TET2, DNMT3A, JAK2, TP53 and PPM1D genes, and cardiovascular, metabolic, and renal pathologies. In some instances, it was shown that the pathological effects of a CHIP driver mutation (TET2, TP53 and PPM1D) could be mitigated with specific anti-inflammatory drugs. Of particular relevance to the proposed studies, there is an accelerated form of clonal hematopoiesis that is observed in individuals that have undergone myelosuppressive treatment and is referred to as "therapy-related clonal hematopoiesis." Under these conditions, it has been shown that there are hematopoietic clonal expansions with a very high frequency of mutations in PPM1D andTP53, both of which are classic DNA damage response genes. In individuals undergoing cytotoxic therapy, the hematopoietic system is likely under extreme stress, and it is thought that mutations in genes such as TP53 and PPM1D confer the mutated hematopoietic stem cell with a survival advantage against genotoxic stress induced by chemotherapy. Recent work from the applicants' laboratories have shown that this form of CHIP can synergize with the genotoxic agent's direct effect on the cardiovascular system to promote a more robust cardiomyopathic phenotype. While the impact of space travel on CHIP is completely unknown, it is reasonable to speculate that space radiation in combination with other space travel-related stresses will lead to radiation-specific and gene-specific accelerations of clonal hematopoiesis. Further, these forms of CHIP may increase the risk of leukemogenic and cardiovascular pathologies in a radiation- and gene-specific manner. There are many forms of genome instability that underpin clonal hematopoiesis including the driver gene mutations described above, mosaic chromosomal alterations, and chromosome aneuploidy – particularly, the loss of a sex chromosome. Loss of the Y chromosome (LOY) in males in the hematopoietic system is the most common post-zygotic mutation in humans and it increases with age. Epidemiological studies show that LOY is associated with all-cause mortality and our LOY mouse models have shortened lifespans and accelerated age-associated diseases (Sano Science 2022). In an unpublished analysis of a single cell data set of highly phenotyped healthy male individuals we were able to computationally assess appreciable leukocyte LOY within the age range of 35 to 55 years old, the typical age of a male astronaut, thus it is likely that male astronauts with significant preexisting LOY undergo spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	While the impact of space travel on CHIP is completely unknown, it is reasonable to speculate that space radiation in combination with other space travel-related stresses will lead to radiation-specific and gene-specific accelerations of clonal hematopoiesis. Further, these forms of CHIP may increase the risk of leukemogenic and cardiovascular pathologies in a radiation- and gene-specific manner.
	The accumulation of somatic mutations is a phenomenon that occurs in all tissues throughout the body during aging. In the hematopoietic system, a somatic mutation can randomly occur in a preleukemic driver gene within a hematopoietic stem and progenitor cell, providing that cell with a competitive advantage and thereby facilitating a clonal expansion. This condition is known as clonal hematopoiesis (CH) and leads to a population of mutant leukocytes in the peripheral blood. Studies suggest that these mutations may alter the immunological profile of leukocytes and can thereby causally contribute to age-related diseases. Thus, understanding how ionizing space radiation supports clone growth and the level at which clones become pathological is of paramount importance. During the current year we have followed and monitored the DNMT3A cohort. This cohort is currently 13 months post radiation exposure. Serial blood sampling has been performed to analyze the expansion of the DNMT3A KO cells, echocardiograms have been performed at two time points, body weights were collected, and survival was tracked. Compared to what we saw in murine models of TP53- and PPM1D-CH, there does not appear to be a radiation effect on DNMT3A-KO cell expansion in white blood cells (WBCs), but the expansion was accelerated in the females versus the males. The DNMT3A-KO cell expansion in the males has lagged behind the females, but the KO cell expansion in the males did increase between the 4-month and 8-month time points. Notably, the analysis of individual leukocyte populations revealed that the DNMT3A mutation predominantly expanded in B cells. The accelerated KO cell expansion in the female cohort did lead to diminished survival. There may be a slight radiation effect on survival in the female cohort. We will continue to follow this DNMT3A cohort until 18 months post radiation. In addition to the DNMT3A cohort, the repeat TP53 wild-type cohort was euthanized, and tissues were harvested in December 2024. With this cohort completed, we have been able to draft more of the manuscript that describes our findings on the models of TP53, PPM1D, and TET2 clonal hematopoiesis. There are many forms of genome instability that underpin clonal hematopoiesis, including the driver gene mutations described above, mosaic chromosomal alterations, and chromosome aneuploidy – particularly, the loss of a sex chromosome. Loss of the Y chromosome (LOY) in males in the hematopoietic system is the most common post-zygotic

Task Progress:	mutation in humans, and it increases with age. Epidemiological studies show that LOY is associated with all-cause mortality, and our LOY mouse models have shortened lifespans and accelerated age-associated diseases (Sano Science 2022). In an unpublished analysis of a single cell data set of highly phenotyped healthy male individuals, we were able to computationally assess appreciable leukocyte LOY within the age range of 35 to 55 years old, the typical age of a male astronaut; thus, it is likely that male astronauts with significant preexisting LOY undergo spaceflight. The effects of space radiation on sex chromosome aneuploidies in leukocytes and their effects on immune function remain largely uninvestigated. Thus, we have undertaken a systematic analysis of the conditions that contribute to hematopoietic LOY in mice. These studies reveal that age, germline and somatic mutations in TP53, and irradiation increase the risk of LOY in mice. Thus, we have submitted a proposal to run the LOY cohort in the Spring 2026 NSRL campaign. We are currently running pilot studies with one of our LOY mouse models to determine the ideal radiation paradigm and dose that will elicit the best phenotype. Of the 10 dosing paradigms, ranging in total doses of 0.5Gy to 10Gy, the 1Gy every other day for 3 doses appears to have the best effect on LOY cells in this mouse strain. We will repeat this paradigm with a cohort of mice that have a baseline blood draw to ensure that there are no outlier mice with high levels of LOY prior to irradiation. In addition to the LOY mouse studies, we have started to examine the single-cell multiomic sequencing datasets of astronauts on particular missions to more fully understand the effects of spaceflight on the sex-specific transcriptional and epigenetic regulatory pathways in XX, XY, and XO leukocytes. To date, we have received the single-cell RNA and ATAC sequencing datasets for immune cell populations from the Inspiration4 crewmembers (2 males, 2 females). Based on the 6 timepoints, the average LOY (mean ± SEM) of the two male astronauts was 2.86 ± 0.76% (C001) and 2.93 ± 0.52% (C004), where LOY was defined as cells with no detectable expression of Y-chromosome genes. Although both male astronauts have detectable levels of LOY, these levels are below the threshold considered to be clinically relevant. In the future, we hope to acquire datasets from other commercial missions (e.g., Polaris Dawn, FRAM2, and Axiom-2–4) as they become available.
Bibliography Type:	Description: (Last Updated: 07/15/2026)
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